

Inferring critical transitions from timeseries

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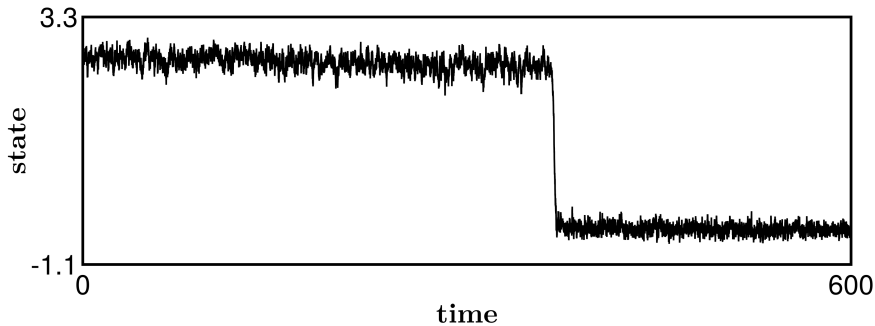
†Auckland Bioengineering Institute



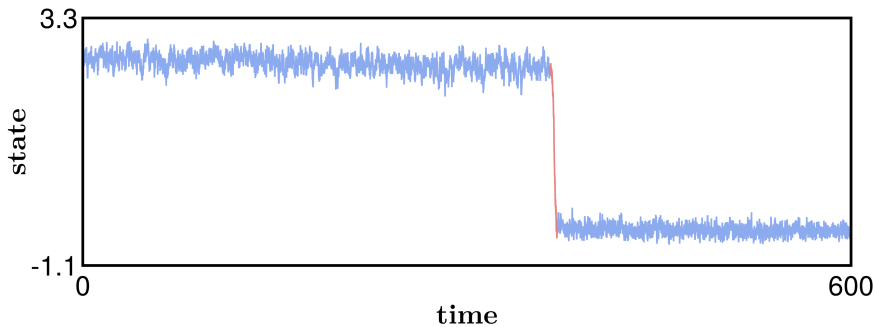
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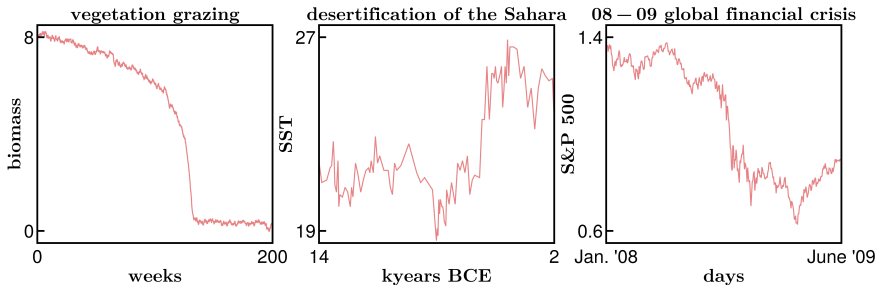
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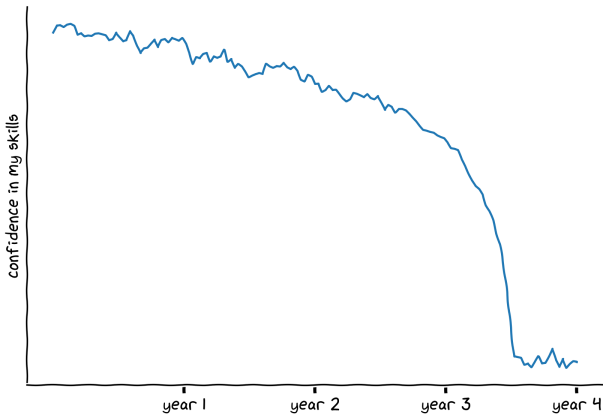


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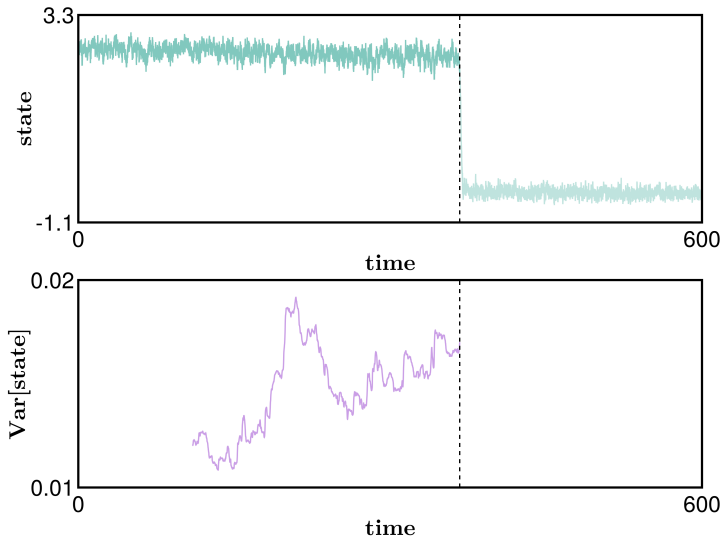


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A PhD's tipping point in imposter syndrome



Inferring critical transitions from timeseries



The first early-warning signal (EWS)

Consider the following Itô process with additive white noise (i.e. $\sigma > 0$)

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which admits a stationary probability density with $\theta = -2\sqrt{-\mu}$ and covariance

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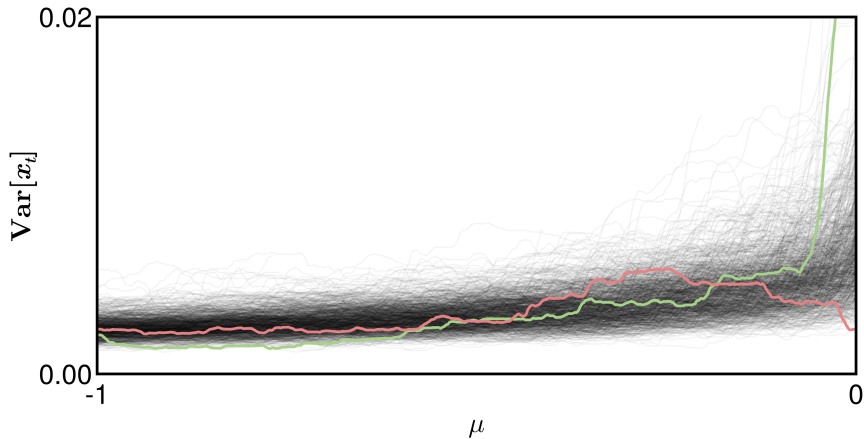
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If we now consider instantaneous time increments (i.e. $t \rightarrow s$) then

$$\text{Cov}[X_s, X_t] \rightarrow \text{Var}[X_t] = \frac{\sigma^2}{2\theta} \rightarrow \begin{cases} 0, & \mu \rightarrow \infty, \\ \infty, & \mu \rightarrow 0. \end{cases}$$

A quest for robust EWSs



Critical slowing down (CSD)

Consider the deterministic limit of the noisy saddle-node normal form in 2-D

$$\begin{cases} \dot{x} = -\mu - x^2, \\ \dot{y} = -\alpha y, \end{cases}$$

with a **stable node** $\mathbf{x}_\mu^{(1)} = [\sqrt{-\mu}, 0]^T$ and a **saddle** $\mathbf{x}_\mu^{(2)} = [-\sqrt{-\mu}, 0]^T$ equilibria.

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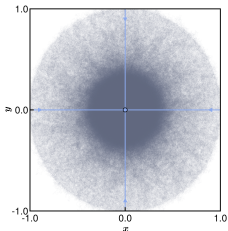
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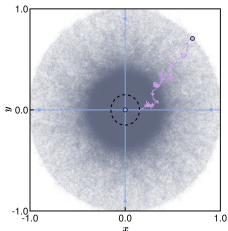
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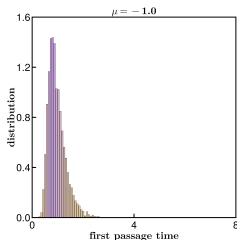
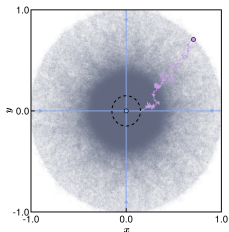
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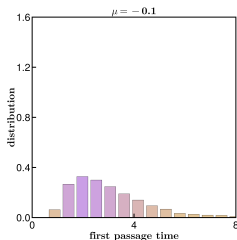
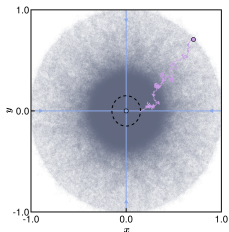
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Large deviation theory

A candidate EWS for capturing CSD

Consider a vector field $f(x; \mu)$ with equilibria $x_{\text{stb}, \text{uns}}$.

If the vector field is conservative then its potential $V(x; \mu)$ has a local minima at $a \equiv x_{\text{stb}}$ and a local maxima at $b \equiv x_{\text{uns}}$.

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Approaching a saddle-node bifurcation ($\mu \rightarrow \mu_{\text{bif}}$) we have $|x_{\text{stb}} - x_{\text{uns}}| = |a - b| \rightarrow 0$, meaning that

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We can use the escape rate as an EWS!

$$r_{a \rightarrow b} = \begin{cases} 0, & \text{if } |\mu - \mu_{\text{bif}}| \rightarrow \infty, \\ 1, & \text{if } |\mu - \mu_{\text{bif}}| \rightarrow 0. \end{cases}$$

Transition probabilities

Consider a conservative system $f(x; \mu) = -\nabla V(x; \mu)$ where V is a scalar potential of the state's variables.

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Let $\mu \in \mathbb{R}$ be fixed over $\Delta t \gg 0$; if $\frac{\Delta p}{\Delta t} \approx 0$ (i.e. I ignore the transient regime) then my particle's transition probability solves the second-order IVP

$$\begin{cases} Dp'' + V''p + V'p' = 0, \\ p'(\pm\infty) = 0. \end{cases}$$

Stationary equilibrium distribution

I can reduce the 1-D (state variable p) second-order ODE to a 2-D (state variables p and p') first-order system of ODEs.

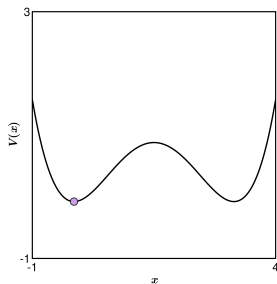
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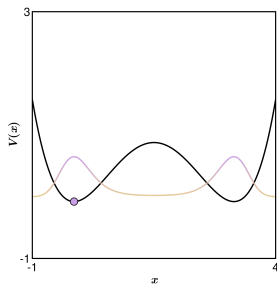


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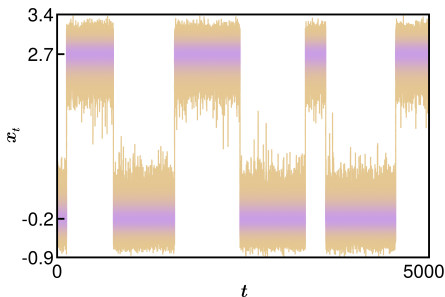
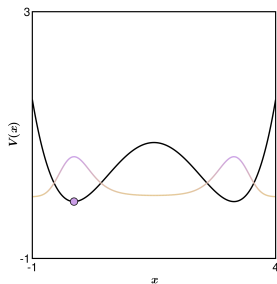


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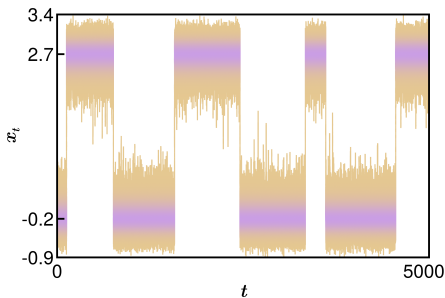
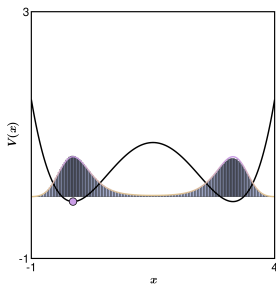


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A numerical method to predict tipping

Input: timeseries x_t over $\Delta t > 0$, threshold > 0

define $V_*(x, \mathbf{c}) = \mathbf{c}[1]x + \mathbf{c}[2]x^2 + \mathbf{c}[3]x^3$;

define $\rho(x, \mathbf{c}) = \exp(-V(x, \mathbf{c}))$;

residuals = detrend(x_t , Δt);

histogram = fit(residuals);

while *error* > *threshold* **do**

 guess = solve_lls($\rho(x)$, histogram) + rand(3);

$\rho(x)$ = normalise($\rho(x, \text{guess})$);

 solution = solve_nlls($\rho(x)$, histogram, guess);

 error = $\int (\rho(x, \text{solution}) - \text{histogram})^2 dx$

a, b = compute_extrema($V_*(x, \text{guess})$);

ews = $\frac{1}{2\pi} \sqrt{|V_*''(b, \text{guess})| V_*''(a, \text{guess})} \exp(-(V_*(b, \text{guess}) - V_*(a, \text{guess})))$;

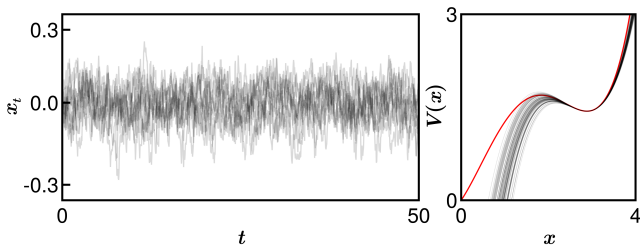
Output: escape rate **ews** from the minima a of V_*

Nonlinear least-squares regression

To characterise the statistical properties of the method let's consider an ensemble of $N \gg 1$ i.i.d. overdamped particles in a cubic potential

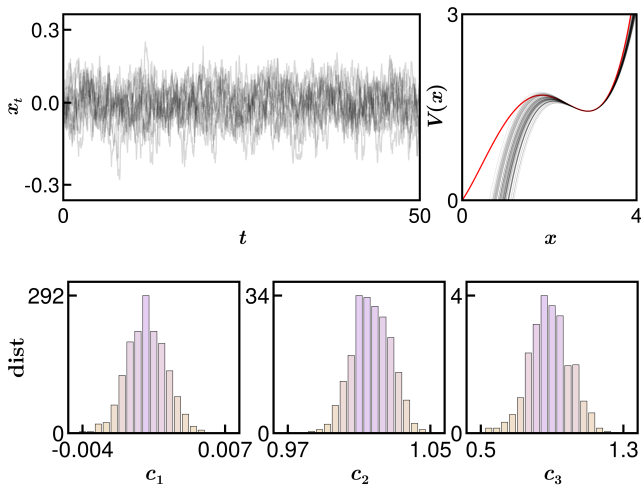
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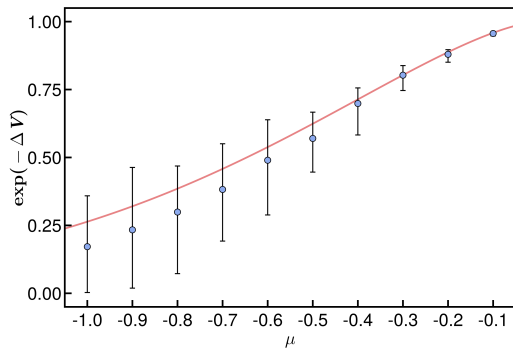


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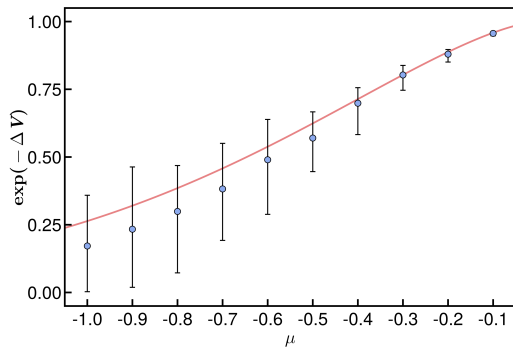
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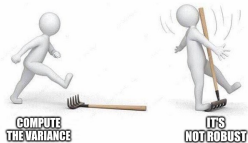
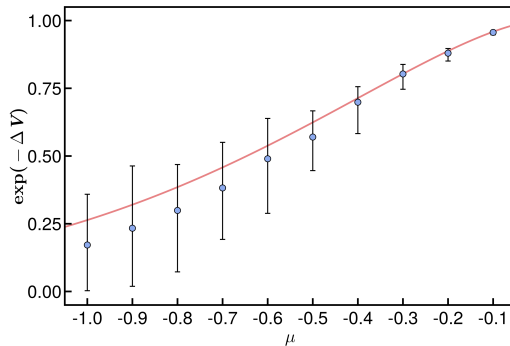


COMPUTE
THE VARIANCE

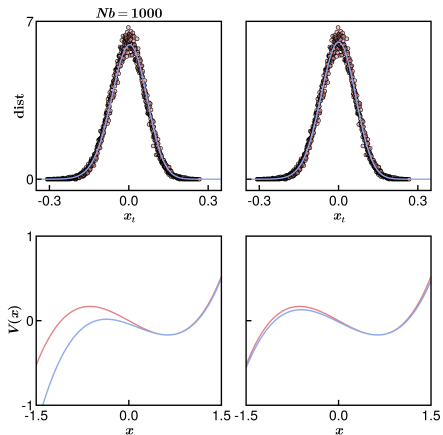


IT'S
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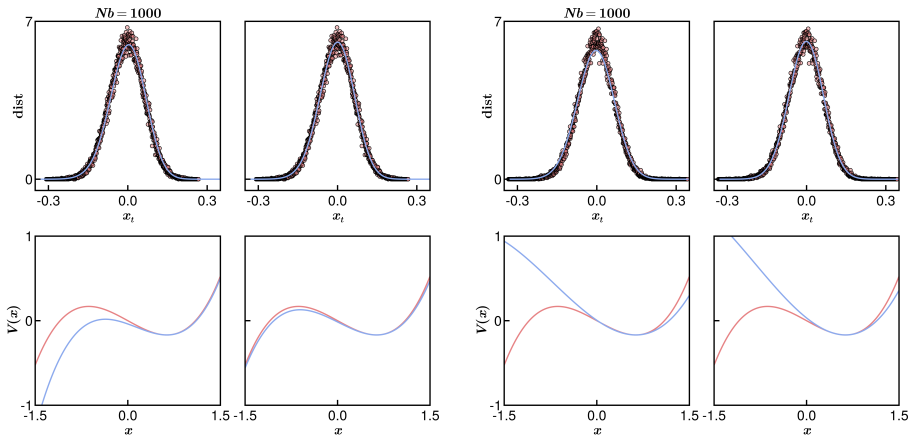
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Investigating the lack of robustness

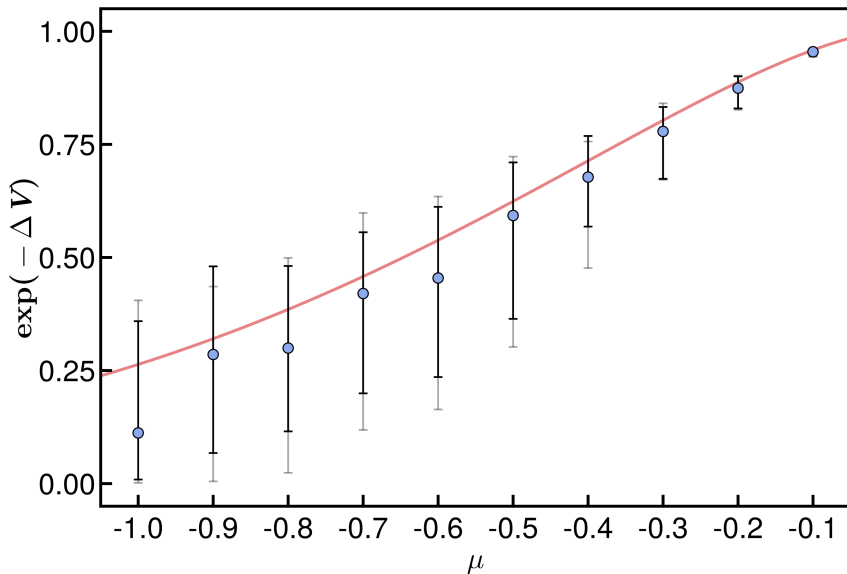


Investigating the lack of robustness



Symmetrically weighted least-squares

First attempts at stabilizing the method



That's the end folks!

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-  P. Ritchie and J. Sieber, Chaos **26** (2016).
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Thank you.

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